
CASE REPORT

Childhood Stroke after Influenza Vaccination

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ABSTRACT

Neurological complications following influenza vaccination are rare in children, with the most common being seizures, and less commonly encephalopathy and Guillain-Barré syndrome. We report here a 10-year-old child who developed cerebellar stroke one day after receiving the seasonal trivalent influenza vaccine; to our knowledge this is the first reported case in literature. The child presented with acute-onset vertiginous giddiness, blurred vision, headache and non-bilious vomiting. Clinical examination revealed marked ataxia with past pointing worse on the right. Laboratory investigations did not reveal any thrombophilic tendency and there was no suggestion of embolic phenomena from a distant site. Magnetic resonance imaging (MRI) of the brain showed two areas of restricted diffusion in the left cerebellar hemisphere. She was commenced on aspirin and underwent rehabilitation before being discharged six days after admission in good health. She recovered completely by outpatient review and aspirin was discontinued eight months after presentation.

Keywords: Children, Influenza vaccine, Stroke, Vaccine adverse events

INTRODUCTION

Neurological complications following an influenza vaccination are rare in children¹. While Guillain-Barré syndrome is the most common serious neurological adverse event reported after trivalent inactivated influenza vaccine in adults, in children, the most commonly reported serious neurological adverse event are seizures (both febrile and afebrile)^{2,3}. Other neurological complications that were reported in children in the above reports were much less common, and include Guillain-Barré syndrome and encephalopathy. We report an unusual case of a child who presented with vertiginous giddiness one day after the seasonal trivalent influenza vaccination (prior to the availability of monovalent influenza A [H1N1] 2009 vaccine) and was found to have cerebellar stroke.

CASE REPORT

A 10-year-old girl of Chinese-Thai descent presented with acute-onset vertiginous giddiness,

blurred vision, headache and non-bilious vomiting. She had new-onset vertiginous giddiness six hours prior to admission, associated with mild frontal headache, which was worse on sitting up. She also complained of a sudden inability to balance herself and had vomited more than 10 times prior to admission. There were no associated symptoms of tinnitus, hearing loss, focal numbness or weakness, and there was no history of head trauma. However, she did receive the influenza vaccine (Fluarix™, GlaxoSmithKline Pte Ltd) one day prior to the onset of symptoms. She had a past medical history of a traumatic left distal tibial fracture two months prior to presentation, and reported mild infrequent headaches in the past but which had not occurred in the preceding six months. There was no past history of vertigo. There was no family history of migraine headaches, myocardial infarcts or other thrombophilic tendency. Her grandfather had a stroke at an advanced age. There was no prior history of miscarriages in her mother. On

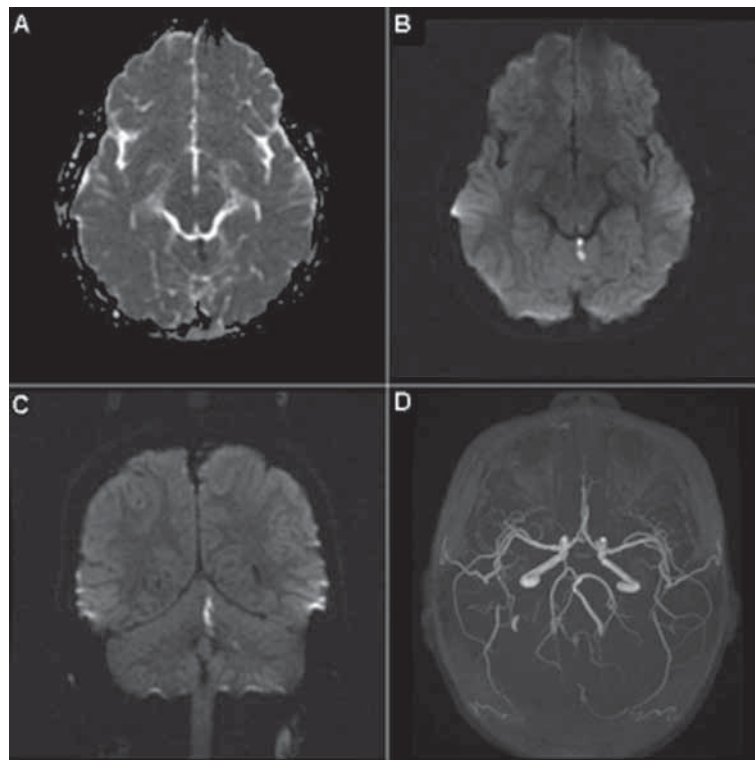


Fig. 1. MRI of cerebellar stroke in a child one day after seasonal trivalent influenza vaccine. Panels: a. with contrast; b. diffusion-weighted image, transverse view; c. diffusion-weighted image, coronal view; d. MRA showing normal vessels apart from a fetal origin of the right posterior cerebral artery.

examination, she was tired but cooperative. Her Glasgow Coma Score was 15/15. There was no meningismus though she held her head tilted to the right. She was able to sit and stand up independently but required some effort to steady herself with her arms, and walked with an ataxic gait (she was unable to perform tandem gait). Her coordination was impaired and there was mild past-pointing, which was more prominent on the right, her peripheral tone, reflexes, power, and sensations were normal. Her visual acuity was 20/20 and her visual fields were intact. Her extra-ocular eye movements were grossly intact but she experienced diplopia on upward gaze. Her pupils reacted normally to direct and consensual light and fundoscopy revealed normal optic discs. Her cardiovascular examination was normal and there were no carotid bruits heard.

Ophthalmologic evaluation showed mild left fourth nerve palsy. MRI of the brain showed two areas of restricted diffusion in the left cerebellar hemisphere, the first in the superior cerebellar vermis just left of the midline at the roof of the fourth ventricle and the second at the postero-superior part of the left cerebellar hemisphere

(Fig.1). Magnetic resonance angiogram (MRA) showed no abnormality of the circle of Willis or vessels apart from a fetal origin of the right posterior cerebral artery.

Laboratory investigations were as follows: C-reactive protein was <5 mg/L, erythrocyte sedimentation rate (ESR) was 26 mm/50 minute (min). Liver function tests and electrolytes were normal (protein 72 g/L, albumin 39 g/L (41-53 g/L), total bilirubin 14 μ mol/L, alkaline phosphatase 328 U/L, alanine transaminase 10U/L, aspartate transaminase 19 U/L, gamma glutamyl tranferase 6 U/L, sodium 136 mmol/L, potassium 4.3 mmol/L, urea 3.5 mmol/L, creatinine 48 μ mol/L). Full blood count was normal (haemoglobin 12.7 g/dL, white cell count 7.2×10^9 /L, platelets 350×10^9 /L). Prothrombin time was 13.3 seconds (sec) (11.7-15.1 sec), international normalised ratio (INR) was 1.04, and activated partial thromboplastin time was 36.1 sec (31.8-43.7 sec). Factor V Leiden was negative, antinuclear antibody, anti-double stranded deoxyribonucleic acid and anti-cardiolipin antibodies were negative. Lupus anticoagulant was not demonstrated. Plasma homocysteine was 8.6 μ mol/L, Protein C activity was 120% (70-140%),

functional Protein S activity was 83% (65-140%) and antithrombin III activity was 111% (80-120%). Anti- β 2 glycoprotein I immunoglobulin M and immunoglobulin G were negative.

She was commenced on aspirin at a dose of 5 mg/kg/day from the second day of admission. She had a right eye patch applied for symptomatic control of diplopia. She also underwent physical and occupational therapy, and became ambulatory by the sixth day of admission. Her clinical condition continued to improve and she was eventually discharged well from hospital on aspirin. At her six month review at the outpatient clinic she had completely recovered with no residual or new neurological symptoms or signs. Repeat blood tests were normal apart from an ESR of 20 mm/50 min. She was tapered off aspirin after eight months of treatment and has remained well at follow-up 10 months after initial presentation.

DISCUSSION

To our knowledge, childhood stroke has not been previously reported in association with the influenza vaccination. A review of literature revealed two recent reports of stroke temporally associated with influenza vaccination in adults; however both reported cases had underlying co-morbidities that substantially increased their predisposition to developing a stroke. The first report was of a 55-year-old Caucasian female with a history of osteoporosis and non-erosive knee arthritis. The patient developed a left middle cerebral artery ischaemic infarct four days after the influenza vaccination. She was subsequently found to have systemic lupus erythematosus with secondary anti-phospholipid syndrome, and the cause of her ischaemic stroke was believed to be due to the development of antibodies to phospholipids, leading to endothelial damage causing in-situ occlusion⁴. The second report occurred in a 64-year-old Caucasian male who had a history of atrial fibrillation on warfarin for stroke prophylaxis⁵. He passed away following a large left temporal, parietal and occipital haemorrhagic infarct 4½ weeks after receiving an inactivated influenza vaccine; at the time of the stroke, he was documented to have a greatly increased INR that had increased from two at the time of vaccination (and stable INR over the previous six months) to >15. It was hypothesised that in some individuals an interaction could occur between warfarin and the influenza vaccine that leads to increased

anticoagulation and a subsequent increased risk of haemorrhagic infarcts. Despite the above reports, other authors have been unable to establish a definite relationship between influenza vaccination and stroke in these patient populations⁶⁻⁸. Our patient had an extensive workup that did not reveal any underlying prothrombotic or autoimmune conditions (her coagulation profiles were normal), and she was not on anticoagulants prior to the development of her ischaemic stroke.

Childhood stroke following varicella vaccination has been previously described, and it was postulated that, similar to natural varicella infection, varicella vaccine might induce cerebral angiopathy with resultant narrowing and beading of the cerebral arteries leading to ischaemic infarcts⁹. However, a subsequent large cohort study examining this relationship failed to identify a significant association¹⁰. Bell et al. recently described a four-year-old child with acute ischaemic stroke following influenza A infection and found partial occlusion of his proximal left middle cerebral artery on an MRI; the authors noted that in the majority of infection-associated strokes, occlusion or narrowing of proximal large cerebral vessels were observed¹¹. Nonetheless, we did not see similar imaging findings of cerebral angiopathy in our patient following an MRA to explain the nature of her ischaemic stroke, although an MRA may not have been sensitive enough to detect small vessel lesions as in small vessel central nervous system vasculitis and functional imaging studies would have been required.

The presence of an association does not imply causality. The burden of influenza (in terms of infection rates, hospitalisations, morbidity and mortality) is significant and this was recently reinforced with the emergence of the pandemic Influenza A (H1N1) 2009; influenza vaccination remains the most effective method of preventing an influenza virus infection and its complications¹². Furthermore, there is increasing evidence that, at least for adults and the elderly, influenza infection itself may be associated with an increased risk of stroke, and that vaccination may reduce that risk¹³⁻¹⁵. The close temporal relationship between an ischaemic stroke in an otherwise healthy 10-year-old child with recent receipt of seasonal influenza vaccination may be entirely coincidental, and does not alter our stance in recommending the influenza vaccination for all children, especially

those at risk of developing complications from an influenza infection.

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